Chapter 5
Surface water flow and soil erosion modeling

The topography of the Earth's surface controls the flow of water and mass over the landscape. Modifications to the surface geometry of the land redirect water and mass flows influencing ecosystems, crop growth, the built environment, and many other phenomena dependent on water and soil. We used Tangible Landscape to explore the relationship between overland flow patterns and landscape topography by manually changing the landscape model while getting near real-time feedback about changing flow patterns. We coupled Tangible Landscape with a sophisticated dam break model to investigate flood scenarios after a dam breach.

5.1 Foundations in flow modeling

Water flow over complex terrain can be described by a bivariate form of the shallow water continuity equation. The continuity equation can be coupled with momentum (Navier-Stokes) equations to simulate flooding due to dam failure.

5.1.1 Overland flow

For shallow water flow the spatial variation in velocity with respect to depth can be neglected and the overland water flow during a rainfall event can be approximated by the following bivariate continuity equation (Julien et al., 1995):

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\,\mathbf{v}) = i_e \tag{5.1}$$

where:

h is the depth of overland flow in m
t is the time in s
$\mathbf{v}$ is the flow velocity vector $\mathbf{v} = (v_x, v_y)$ in m/s
i_e is the rainfall excess (rainfall − infiltration − vegetation intercept) in m/s.

If we assume that the dynamic friction slope of the water surface can be approximated by the static bare ground slope the flow velocity can be estimated by the Manning's equation:

$$\mathbf{v} = \frac{1}{n} h^{2/3} |\mathbf{s}|^{1/2} \mathbf{s_0} \tag{5.2}$$

where:

n is the Manning's coefficient expressed here with units $\mathrm{s\,m^{-\frac{1}{3}}}$
$\mathbf{s}$ is the negative elevation gradient $-\nabla z = -(f_x, f_y)$ of the bare ground surface
$|\mathbf{s}|$ is the magnitude of elevation gradient (slope, see subsection 4.1.3)
$\mathbf{s_0}$ is the unit vector in the flow direction (the negative elevation gradient direction).

For steady state water flow and a steady rainfall excess rate the continuity equation has the following form:

$$\partial h / \partial t = 0 \quad \longrightarrow \quad \nabla \cdot (h\ \mathbf{v}) = i_e \tag{5.3}$$

We introduce a diffusion-like term proportional to $\nabla^2[h^{5/3}]$ to incorporate the diffusive wave effects in at least an approximate way. The spatially distributed, steady overland water flow is then expressed as follows:

$$-\frac{\varepsilon}{2} \nabla^2 [h^{5/3})] + \nabla \cdot (h\ \mathbf{v}) = i_e \tag{5.4}$$

where ε is a spatially variable diffusion coefficient. The diffusion term, which depends on $h^{5/3}$ instead of h, makes the equation (Eq. 5.4) linear in the function $h^{5/3}$ which enables us to solve it by the Green's function method using a stochastic technique referred to as path sampling method (Mitasova et al., 2004).

The path sampling method is based on the duality between the particle and field representation of spatially distributed phenomena. In this concept the density of particles in a space defines a field and vice versa, i.e. a field is represented by particles with a corresponding spatial distribution. Using this duality, processes can be modeled as the evolution of fields or the evolution of spatially distributed particles (Mitasova et al., 2004).

The flow evolution (accumulation process) can be also interpreted as an approximation of a dynamical solution for shallow water flow, in which velocity is mostly controlled by terrain slope and surface roughness rather than water depth and friction slope. Therefore the change of velocity over time at a given location is negligible. The robustness of the path sampling method enables us to simulate complex, spatially variable conditions and efficiently explore

flow patterns for a wide range of landscape configurations. This method is implemented in GRASS GIS as the module *r.sim.water*.

5.1.2 Dam break modeling

Surface water flow and flooding due to a dam breaking are also represented by the shallow water flow equation, but in this application we assume that there is no rainfall and the only source is water from the reservoir. In order to ensure a realistic representation given a large mass of moving water the shallow water continuity equation:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\,\mathbf{v}) = 0 \tag{5.5}$$

is coupled with the momentum (Navier-Stokes) equation of fluid motion (Cannata and Marzocchi, 2012):

$$\frac{\partial h v_x}{\partial t} + \frac{\partial h v_x^2}{\partial x} + \frac{\partial h v_x v_y}{\partial y} = S_x \tag{5.6}$$

$$\frac{\partial h v_y}{\partial t} + \frac{\partial h v_y v_x}{\partial x} + \frac{\partial h v_y^2}{\partial y} = S_y \tag{5.7}$$

where $\mathbf{S} = (S_x, S_y)$ is the source vector (water flowing from the lake):

$$S_x = -gh\left(\frac{\partial z_w}{\partial x} + \frac{n^2 v_x |\mathbf{v}|}{h^{4/3}}\right) \qquad S_y = -gh\left(\frac{\partial z_w}{\partial y} + \frac{n^2 v_y |\mathbf{v}|}{h^{4/3}}\right) \tag{5.8}$$

and

- g is the gravitational acceleration in m/s^2
- z_w is the water level expressed as elevation above sea level in m
- n is the Manning's roughness coefficient expressed here with units m$^{-\frac{1}{3}}$s
- $|\mathbf{v}|$ is the flow velocity magnitude $|\mathbf{v}| = (v_x^2 + v_y^2)^{1/2}$ in m/s

The resulting partial differential equations are hyperbolic and non-linear and therefore must be solved numerically. The finite volume method with an upwind conservative scheme proposed by Ying et al. (2004) is used to solve the equations on a regular grid which makes the model suitable to GIS-based implementation. Green's theorem is used to obtain the discrete equations. The numerical method explicitly solves the governing equations in two separate steps. First, the continuity equation 5.5 is evaluated deriving the water depth at time $t + \Delta t$ and later on these values are used in the source term (Eq. 5.8) to solve the momentum equations 5.6,5.7 and evaluate the flow velocities at time $t + \Delta t$. More details about the numerical solution can be found in Cannata and Marzocchi (2012). The model is implemented in GRASS GIS as an add-on module *r.damflood*.

5.1.3 Sediment transport, erosion, and deposition

The *Simplified Erosion/Deposition model* (*SED*) model estimates sediment transport across a complex landscape and the resulting pattern of erosion and deposition using the idea originally proposed by Moore and Burch (1986). The model assumes that the sediment flow can be approximated by sediment transport capacity and the net erosion and deposition is transport capacity limited. It combines the Universal Soil Loss Equation parameters and upslope contributing area per unit width A to estimate the sediment flow T:

$$T \approx RKCPA^m(\sin\beta)^n \tag{5.9}$$

where:

- T is the sediment flow rate in kg m^{-1} s^{-1}
- R is the rainfall factor in MJ mm (ha hr yr)$^{-1}$
- K is the soil erodibility factor in ton ha hr (ha MJ mm)$^{-1}$
- A is the upslope contributing area per unit width in m
- β is the steepest slope angle
- C is the dimensionless land-cover factor
- P is the dimensionless prevention-measures factor.

The net erosion/deposition D in kg m^{-2}s^{-1} is then computed as a divergence of sediment flow:

$$D = \nabla \cdot (T\mathbf{s_0}) = \frac{\partial(T\cos\alpha)}{\partial x} + \frac{\partial(T\sin\alpha)}{\partial y} \tag{5.10}$$

where $\mathbf{s_0} = (\cos\alpha, \sin\alpha)$ is the unit vector of the steepest slope (flow) direction (direction of the negative gradient $-\nabla z$) given by the aspect angle α (see Section 4.1.3). The exponents m, n control the relative influence of the water and slope terms and reflect the impact of different types of flow. The typical range of values is $m = 1.0-1.6$, $n = 1.0-1.3$ with the higher values reflecting the pattern for prevailing rill erosion with more turbulent flow when erosion sharply increases with the amount of water. Lower exponent values close to $m = n = 1$ better reflect the pattern of the compounded, long term impact of both rill and sheet erosion and averaging over a long term sequence of large and small events.

5.2 Case study: the hydrological impact of development

The development on North Carolina State University's Centennial Campus raises concerns about the impact of stormwater runoff and erosion during construction. We used Tangible Landscape to explore changes in overland

flow and erosion patterns caused by potential construction site grading and to provide feedback for the design of stormwater and erosion control measures.

In this case study we worked with the same area and physical model used in Section 4.2.1. To familiarize ourselves with the basic hydrologic conditions of this site we computed flow accumulation and delineated watershed boundaries. First, we set the computational region to match our study area and the resolution to 1 m and then we ran the least cost path flow tracing implemented in the module *r.watershed* to derive the flow accumulation raster and watersheds. We set the threshold for the approximate size of the watershed areas to 1000 grid cells and we converted the areas to a vector representation so that we could display the watershed boundaries:

```
g.region n=223765 s=223015 e=638450 w=637700 res=1
r.watershed elevation=elevation accumulation=flow_accum \
    basin=watersheds threshold=1000
r.to.vect input=watersheds output=watersheds type=area
```

The resulting map that combines the flow accumulation raster with the vector representation of watershed boundaries and contours was then projected over the model in Fig. 5.1a.

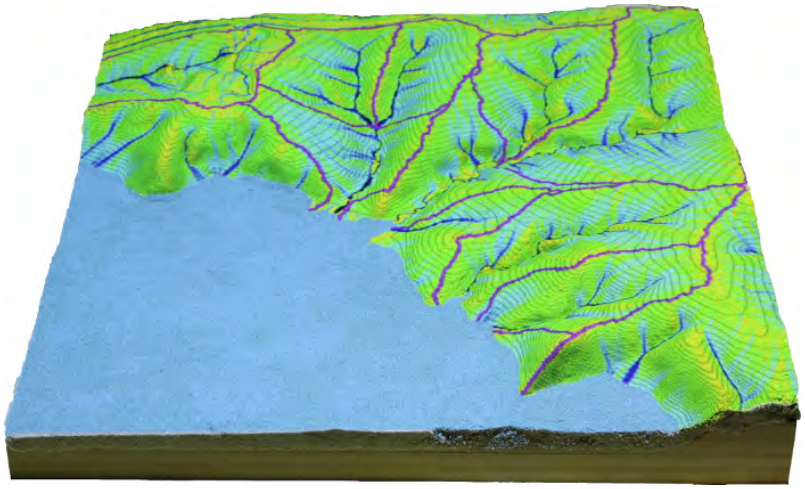
5.2.1 The impact of development on surface water flow

To explore the potential impact of new development we graded sites for new apartment buildings with flat slopes and constructed an access road (see Fig. 4.7d). We carved a narrow culvert to allow water to flow under the road. By calculating flow accumulation and watersheds for the modified topography (Fig. 5.1b) we could immediately see that the watershed boundaries in the developed area changed significantly due primarily to the new road acting as an artificial watershed boundary.

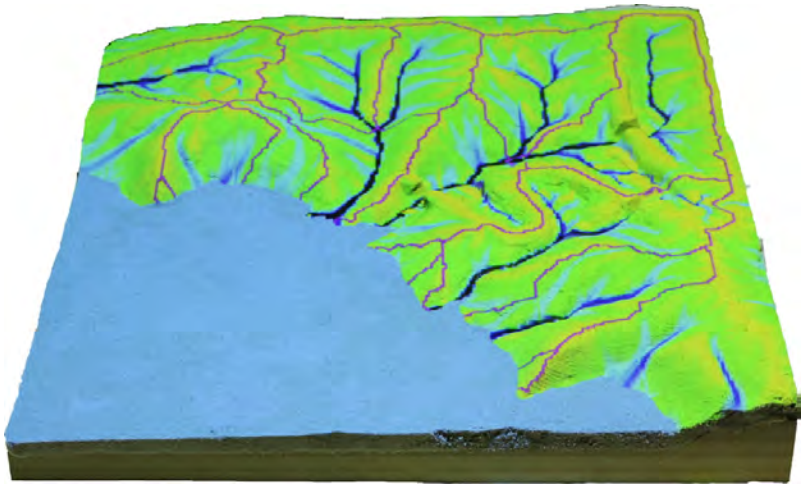
Flow accumulation derived by the module *r.watershed* is based on the least cost path algorithm and is not designed to represent water depth and pooling of water in depressions. Therefore, we used the shallow water flow model implemented in the module *r.sim.water* to simulate the overland flow depth during a storm event for the current conditions and after the grading. We first derived the components of elevation surface gradient $\nabla z = (f_x, f_x)$ (parameters dx, dy in the module *r.slope.aspect*) and then ran the water flow simulation with the rainfall excess value of 150 mm/hr:

```
r.slope.aspect elevation=scan dx=scan_dx dy=scan_dy
r.sim.water elevation=scan dx=scan_dx dy=scan_dy \
    rain_value=150 depth=depth
```

The resulting water depth maps were projected over the model with the initial terrain conditions and then in near real time as the terrain was modified with the final water depth pattern shown in Fig. 5.2b. Note that we have used an extremely high value of rainfall excess to provide rapid maps of water depth



(a)



(b)

Fig. 5.1: Flow accumulation and watershed boundaries computed with *r.watershed* before and after the terrain modifications shown in Fig. 4.7.

as the model was modified. Once we have the final design we can run the simulation with a selected storm and obtain realistic water depth estimates.

We can also run the module *r.sim.water* in a dynamic mode (-*t* flag) and project the evolution of water depth during the storm event as an animation using the GRASS GIS animation tool.

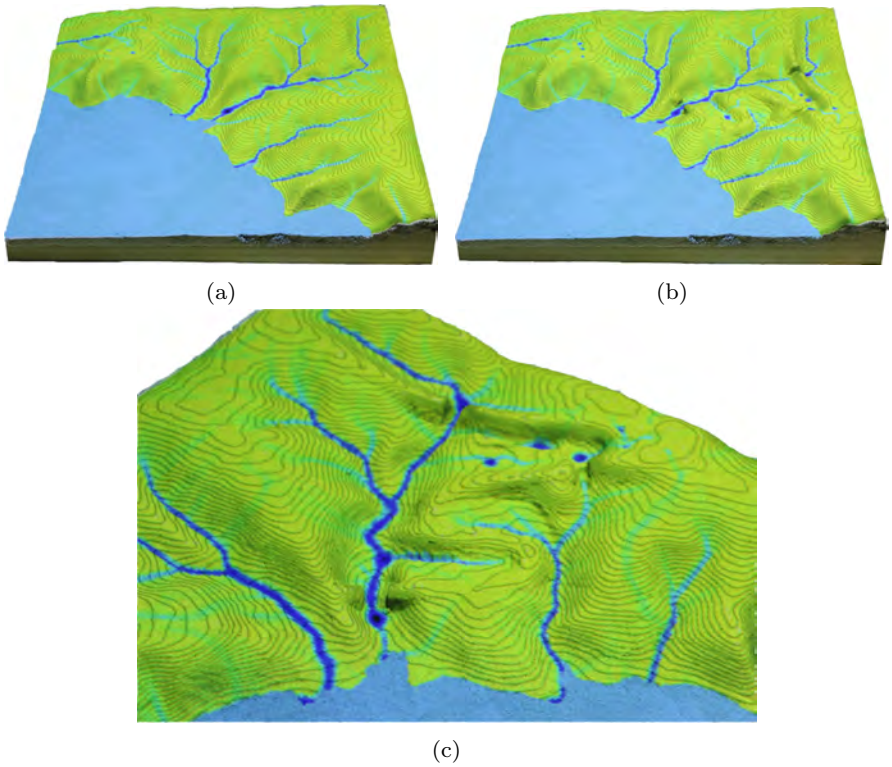


Fig. 5.2: Water depth computed with *r.sim.water* (a) before and (b) after the terrain modifications shown in Fig. 4.7. A detailed view of the modified water flow is shown in (c).

5.2.2 The impact of grading on erosion and deposition pattern

We also explored the impact of grading on sediment transport and erosion/deposition patterns using the SED model. We assumed a uniform rainfall factor $R = 270$ in (hundreds of ft-tonf.in)/(acre.hr.year) as well as uniform soils and land cover with the soil erodibility factor $K = 0.2$ and the land cover factor $C = 0.01$ (grass, no trees). The topographic factor exponents were set to $m = n = 1$ to represent prevailing sheet flow. Since we used uniform values for the R-, K- and C-factors, the pattern of erosion and deposition depends solely upon the slope of the terrain and the upslope contributing area. In the following workflow we first computed the slope and aspect maps of the scanned model. Then we computed flow accumulation and combined it with the slope using map algebra to estimate the topographic factor and sediment transport capacity (see Eq. 5.9).[1] Finally, we computed the divergence of the sediment flow vector field to generate the erosion-deposition map:

```
r.slope.aspect elevation=scan aspect=scan_aspect \
    slope=scan_slope
r.flow elevation=scan flowaccumulation=scan_flowacc
r.mapcalc "scan_sflowtopo = 270. * 0.2 * 0.01 * scan_flowacc * \
    sin(scan_slope)"
r.divergence magnitude=scan_sedflow direction=scan_aspect \
    output=scan_usped
```

The resulting erosion-deposition map is projected over the model in Fig. 5.3.

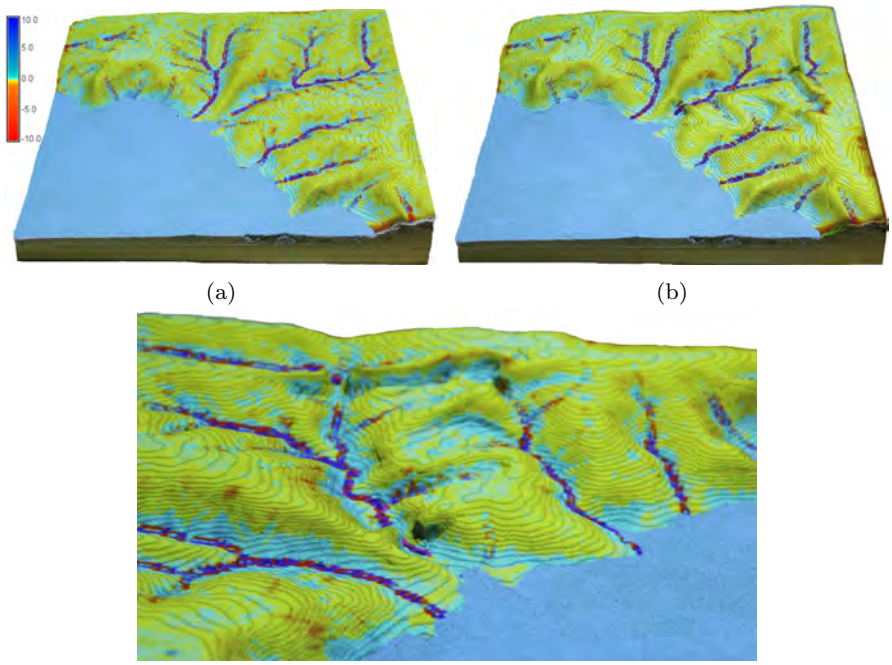
5.3 Case study: dam breach

The Lake Raleigh dam located within our Centennial Campus study area broke in September of 1996 after Hurricane Fran. In this case study we simulated how flood water would spread across the current landscape if the dam was breached again. We also modified the terrain to explore how different morphologies would influence the spread of flooding. We used the add-on *r.damflood*, which was designed specifically to simulate dam breaks and analyze the subsequent flooding.

5.3.1 Site description and input data processing

Fig. 5.4 shows the terrain and the most recent orthophotograph of our Lake Raleigh study area including the dam. The stream flowing out from the lake

[1] For a general case when m, n and resolution are not set to 1 see the erosion modeling tutorial: `http://ncsu-osgeorel.github.io/erosion-modeling-tutorial/grassgis.html`



(a) (b)

(c)

Fig. 5.3: Soil erosion and deposition maps draped over the model (a) before and (b) after grading shown in Fig. 4.7. Detailed view of the erosion-deposition pattern is in (c).

merges with the stream coming from the south in the area just below the dam, which is actively used for recreation providing space for disc golf, running and cycling (Fig. 5.5a).

The dam break simulation requires several input layers including the elevation raster that describes the lake bathymetry and the surrounding topography, the depth of the lake water, the Manning's roughness coefficient, and the geometry of the dam break. The elevation raster was created by fusing the latest DEM based on the 2014 lidar survey with a DEM from the 2001 survey which we used as a proxy for the lake bathymetry. The 2001 survey captured the topography while the lake was drained after the dam failure in 1996 and we considered the 2001 DEM in the lake area a suitable approximation of the current lake bathymetry.

The lake depth raster was computed with the module *r.lake* using the fused DEM as an input. This module fills an area with water for a designated elevation level starting from a given seed point and outputs a map of the water depth for the flooded area. We determined that the current water level elevation for the Lake Raleigh was 85 m by querying the DEM along the

Fig. 5.4: An overview of Lake Raleigh's surroundings with the dam highlighted in red. The digital elevation model on which the orthophoto is draped was created by fusing the latest 2014 data with the lake bathymetry data from 2001.

outline of the lake. We selected an arbitrary point inside the lake area as a seed point. While *r.lake* assigns null values to the cells outside of the filled area, *r.damflood* requires zero values outside of the lake, so we converted null values to zeros with the *r.null* module:

```
g.region n=224026 s=223007 e=639480 w=637798 res=3
r.lake elevation=lake_bottom_dem water_level=85 \
    lake=lake_depth coordinates=638792,223659
r.null map=lake_bottom_dem null=0
```

The Manning's roughness coefficient influences the speed of water. It depends on landcover and varies between 0 and 1 with higher values resulting in slower flows. To derive Manning's coefficients we could for example reclassify a landcover map. In our example for simplicity's sake we created a raster map with a uniform Manning's value:

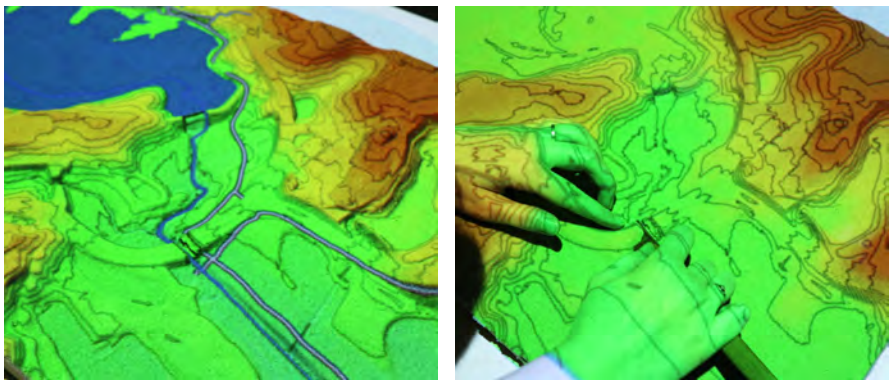
```
r.mapcalc "manning = 0.01"
```

Finally, the module *r.damflood* requires a dam breach raster which represents the height of the breach from the top of the dam. To obtain this raster we scanned the model with the dam. Then we carved a breach into the dam and rescanned the model. We computed the difference using the function defined in code snippet 4.2.2. We used a vertical limit of 1 m (since the scans are already georeferenced) to avoid detecting differences due to noise from scanning:

```
adjust_scan('scan_before', 'scan_after', 'scan_adjusted')
gscript.mapcalc(exp="breach = if(scan_before - scan_adjusted >
    1, scan_before - scan_adjusted, 0)")
```

5.3.2 The impact of the road on flooding

For our first flood simulation we used the current conditions. A major road with a culvert crosses the stream below the dam. We carved a culvert into the sand model by removing a narrow channel of sand from the road and scanned this model to provide input for our simulation.



(a) (b)

Fig. 5.5: (a) The current conditions with a stream and a greenway below the dam; (b) a scenario in which the road has been removed.

Since our sand model does not incorporate the lake bathymetry we had to combine the fused DEM that includes the bathymetry with the scan. We then ran the *r.damflood* module with the input map layers and set the simulation parameters such as the length of the simulation, the time step for creating output maps, and the computational time step. Since the simulation is computationally intensive it may take several minutes to compute depending on the settings and resolution.

```
r.mapcalc "scan_combined = if(lake_mask > 0, lake_bottom_dem, \
    scan_adjusted)"
r.damflood elev=scan_combined lake=lake_depth dambreak=breach \
    manning=manning h=flood timestep=0.1 tstop=1000 deltat=10
```

The series of output raster maps generated by the simulation can be registered in the GRASS GIS Temporal Framework and then visualized using the GRASS GIS Animation Tool. The results for the current conditions are shown in Fig. 5.6. Clearly, the road creates a significant obstacle for the spread of water and causes pooling below the dam. Contained by the road the water floods upstream areas to the west.

In the next simulation we were interested in the conditions before the road was built in 2004. We simply removed the road (Fig. 5.5b) and then rescanned and recomputed the simulation (Fig. 5.7). Removing the road changes the

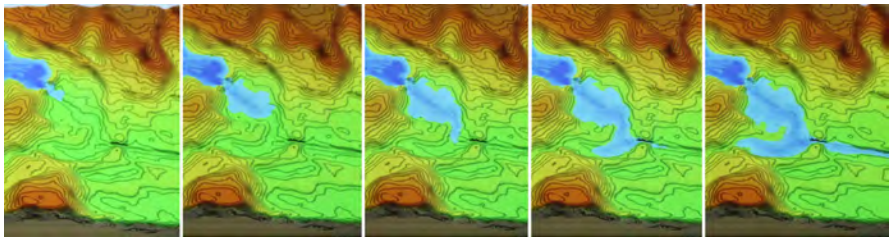


Fig. 5.6: The flood simulation with current conditions

flow pattern and as a result water spreads freely at lower depth thus impacting a larger area, but potentially reducing risk.

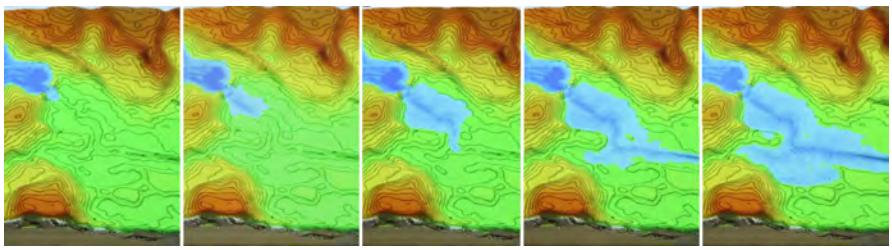


Fig. 5.7: The flood simulation after the road has been removed

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